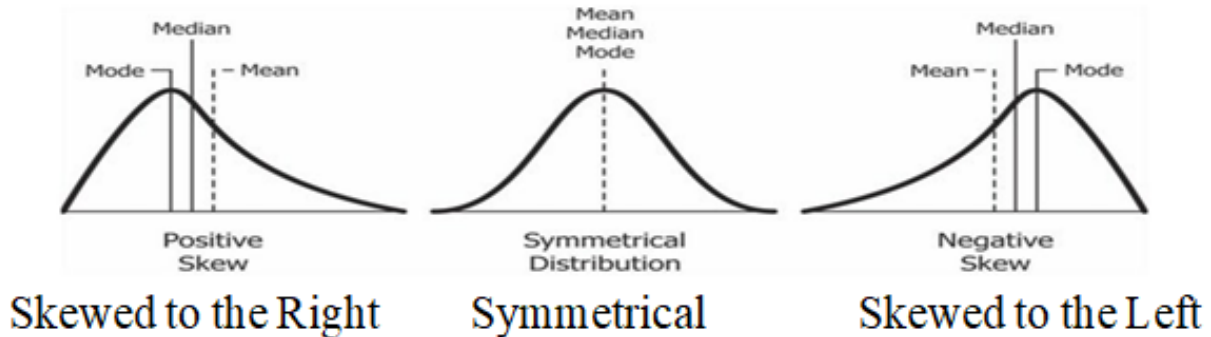


Skewness



- If $\text{Mean} > \text{Mode}$, the skewness is **positive**.
- If $\text{Mean} < \text{Mode}$, the skewness is **negative**.
- If $\text{Mean} = \text{Mode}$, the skewness is **zero**

Moments

Moments are a set of statistical parameters to measure a distribution. Four moments are commonly used:

- **1st moment** - m_1 = Mean (describes central value)
- **2nd moment** - m_2 = Variance (describes dispersion)
- **3rd moment** - m_3 = Skewness (describes asymmetry)
- **4th moment** - m_4 = Kurtosis (describes peakedness)

General formula for moment about mean are given below:

$$m_r = \frac{\sum (x_i - \bar{x})^r}{n}, \text{ where } r=1, 2, 3, 4.$$

$$\text{Put } r=1 \quad m_1 = \frac{\sum (x_i - \bar{x})}{n}, \quad \text{put } r=2 \quad m_2 = \frac{\sum (x_i - \bar{x})^2}{n}, \quad \text{put } r=3$$

$$m_3 = \frac{\sum (x_i - \bar{x})^3}{n} \quad \text{and put } r=4$$

$$m_4 = \frac{\sum (x_i - \bar{x})^4}{n}$$

Example Calculate the first four moments about the mean for the following set of examination marks: 45, 32, 37, 46, 39, 36, 41, 48 & 36

$$\bar{x} = \frac{\sum x_i}{n} = \frac{360}{9} = 40$$

x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$	$(x_i - \bar{x})^3$	$(x_i - \bar{x})^4$
32	-8	64	-512	4096
36	-4	16	-64	256
36	-4	16	-64	256
37	-3	09	-27	81
39	-1	01	-1	1
41	1	01	1	1
45	5	25	125	625
46	6	36	216	1296
48	8	64	512	4096
$\sum x_i = 360$	$\sum (x_i - \bar{x}) = 0$	$\sum (x_i - \bar{x})^2 = 232$	$\sum (x_i - \bar{x})^3 = 186$	$\sum (x_i - \bar{x})^4 = 10708$

$$m_1 = \frac{\sum (x_i - \bar{x})}{n} = 0$$

$$m_2 = \frac{\sum (x_i - \bar{x})^2}{n} = \frac{232}{9} = 25.78 \text{ (marks)}^2$$

$$m_3 = \frac{\sum (x_i - \bar{x})^3}{n} = \frac{186}{9} = 20.67 \text{ (marks)}^3$$

$$m_4 = \frac{\sum (x_i - \bar{x})^4}{n} = \frac{10708}{9} = 1189.78 \text{ (marks)}^4$$

For grouped data

$$m_1 = \frac{\sum f_i (x_i - \bar{x})}{\sum f_i}, \quad m_2 = \frac{\sum f_i (x_i - \bar{x})^2}{\sum f_i}, \quad m_3 = \frac{\sum f_i (x_i - \bar{x})^3}{\sum f_i} \quad \text{and}$$

$$m_4 = \frac{\sum f_i (x_i - \bar{x})^4}{\sum f_i}$$

► **Illustration 4 (Grouped data)** : For the following frequency distribution, find first four central moment.

xi	1	3	5	7	9
fi	2	5	4	7	1

► **Solution:**

xi	fi	fixi
1	2	2
3	5	15
5	4	20
7	7	49
9	1	9
	N = 19	95

Mean $\bar{x} = \frac{\sum f_i x_i}{N} = 95/19 = 5$

xi	fi	fixi	(xi - $\bar{x}$)	fi(xi - $\bar{x}$)	(xi - $\bar{x}$) ²	fi(xi - $\bar{x}$) ²	(xi - $\bar{x}$) ³	fi(xi - $\bar{x}$) ³	fi(xi - $\bar{x}$) ⁴
1	2	2	-4	-8	16	32	-64	-128	512
3	5	15	-2	-10	4	20	-8	-40	80
5	4	20	0	0	0	0	0	0	0
7	7	49	2	14	4	28	8	56	112
9	1	9	4	4	16	16	64	64	256
	N = 19	95	0	0	40	96	0	-48	960

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$$\text{First central moment} = m_1 = \frac{\sum f(x_i - \bar{x})}{\sum f_i} = \frac{0}{19} = 0$$

$$2^{\text{nd}} \text{ central moment} = m_2 = \frac{\sum f(x_i - \bar{x})^2}{\sum f_i} = \frac{96}{19} = 5.052$$

$$3^{\text{rd}} \text{ central moment} = m_3 = \frac{\sum f(x_i - \bar{x})^3}{\sum f_i} = \frac{-48}{19} = 2.526$$

$$4^{\text{th}} \text{ central moment} = m_4 = \frac{\sum f(x_i - \bar{x})^4}{\sum f_i} = \frac{960}{19} = 50.52$$

How to find coefficient of Skewness

Formula

The moment coefficient of skewness β_1 is defined as

$$\beta_1 = \frac{m_3^2}{m_2^3}$$

Question:-

Find coefficient of skewness in example 4

From above example we know that

$$3^{\text{rd}} \text{ central moment} = m_3 = \frac{\sum f(x_i - \bar{x})^3}{\sum f_i} = \frac{-48}{19} = 2.526$$

$$2^{\text{nd}} \text{ central moment} = m_2 = \frac{\sum f(x_i - \bar{x})^2}{\sum f_i} = \frac{96}{19} = 5.052$$

Coefficient of skewness is given by

$$\begin{aligned}\beta_1 &= \frac{m_3^2}{m_2^3} \\ &= \frac{(2.526)^2}{(5.052)^3} = 0.049 > 0\end{aligned}$$

$$\beta_1 > 0$$

it is positive skewed

Kurtosis

Kurtosis refers to the degree of peakedness of a frequency curve. It tells how tall and sharp the central peak is, relative to a standard bell curve of a distribution.

Kurtosis can be described in the following ways:

- **Platykurtic**– When the kurtosis < 0 , the frequencies throughout the curve are closer to be equal (i.e., the curve is more flat and wide)

- **Leptokurtic**– When the kurtosis > 0 , there are high frequencies in only a small part of the curve (i.e., the curve is more peaked)
- **Mesokurtic**- When the kurtosis $= 0$

Description:

1. If $\beta_2 = 3$ or $\gamma_2 = 0$, then curve is said to be mesokurtic;
2. If $\beta_2 < 3$ or $\gamma_2 < 0$, then curve is said to be platykurtic;
3. If $\beta_2 > 3$ or $\gamma_2 > 0$, then curve is said to be leptokurtic;

Question:-

Find kurtosis of grouped and ungrouped data

Ungrouped Data

(From slide 5)

$$\beta_2 = \frac{m_4}{(m_2)^2} = \frac{1189.78}{(25.78)^2} = 1.7902 (\text{platy kurtic} < 3)$$